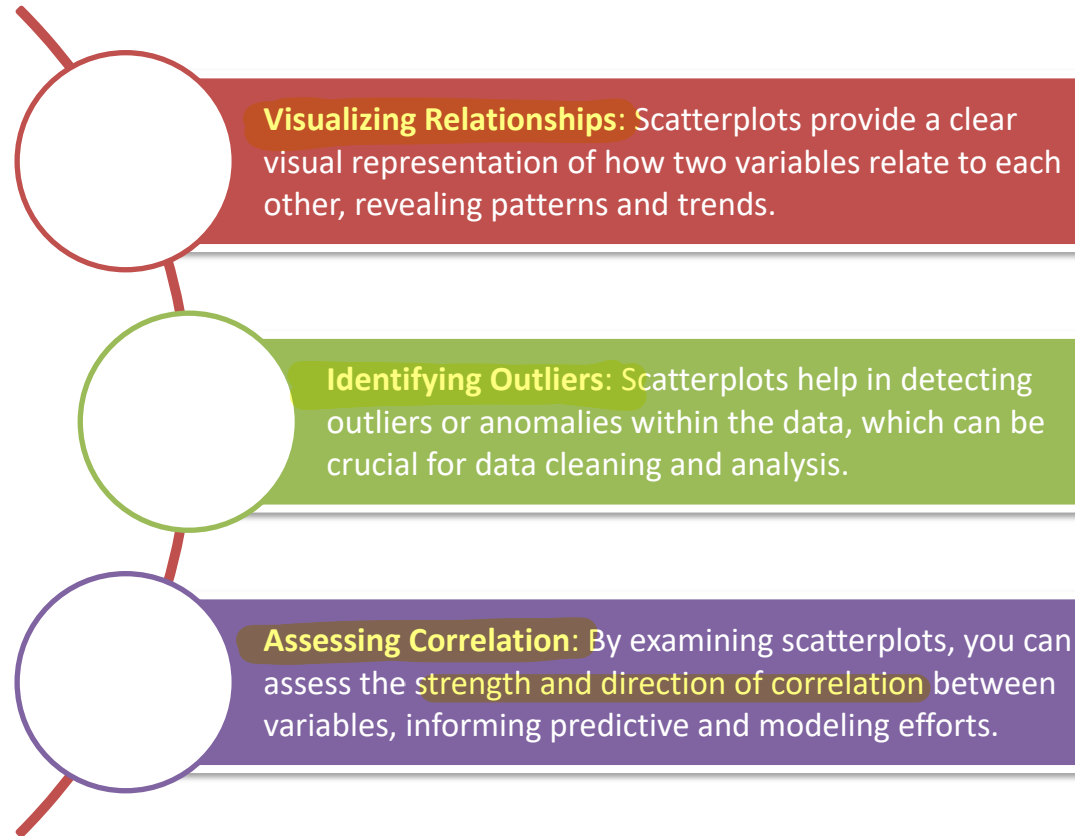


Stat 3113: Lesson 32

Beginning analysis of relationship between two variables - Scatterplots

Motivations



Analysis of
two
variables
together -
Bivariate

We will be trying to
see how two variables
relate to one another

We will focus mainly
on two quantitative
variables

Roles of variables: Often the two variables have a specific role in the analysis

Response Variable (or Dependent Variable) -->y

This is the variable you want to explain variation in, predict, or optimize.

This variable is really the one you are studying.

Your goal regards explanation in some way of this variable.

Explanatory Variable (or Independent or Predictor Variable) -->x

This is a variable you hope explains variation in the response or helps predict the response.

If you know the value of this variable for an individual or object in your study, it will help explain why the response values aren't always the same.

Examples:

Explanatory

1. Cell phone talk time —————→
2. Mutual fund 1st year returns →
3. Time to print one page of text (for printer) —————→

Response

1. Price of cell phone
2. Mutual fund 5th year returns
3. Price of printer

Notes:

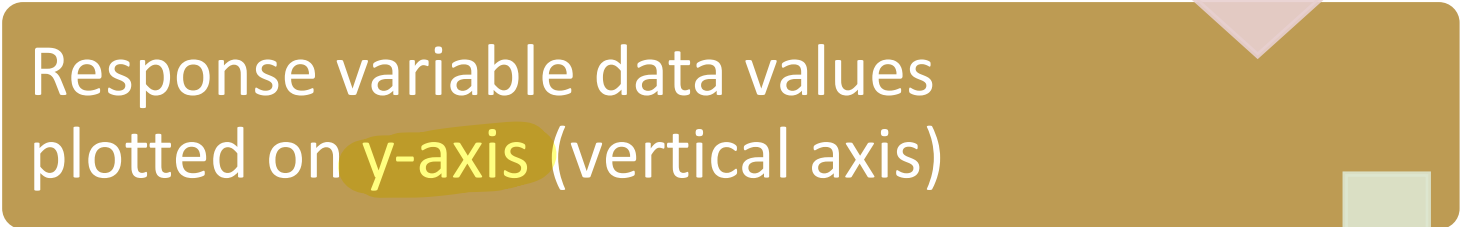
1. Not all bivariate studies have a explanatory/response relationship
2. Our goal is explanation, not necessarily causation

Scatterplot (Scatter Diagram)

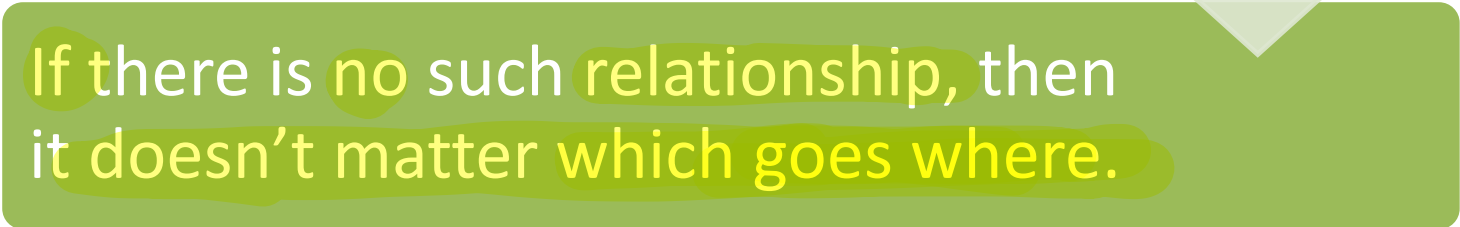
Explanatory variable data values
plotted **on x-axis** (horizontal axis)



Response variable data values
plotted on **y-axis** (vertical axis)



If there is **no** such **relationship**, then
it doesn't matter which goes where.



Things to look for (4 aspects):

1. Outliers
2. Clusters (groupings in the data)
3. Association

Is it negative association (negative sloping)?

Is it positive association (positive sloping)?

Is there no overall association?

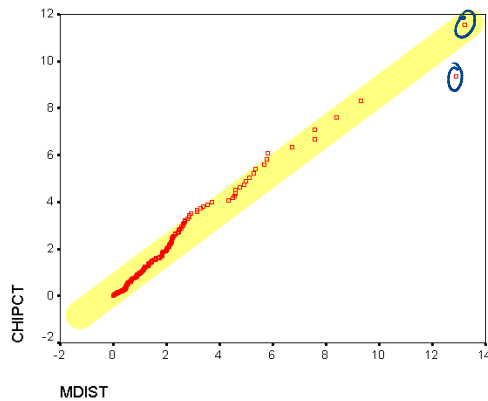
4. Functional Form

Does the data look linear?

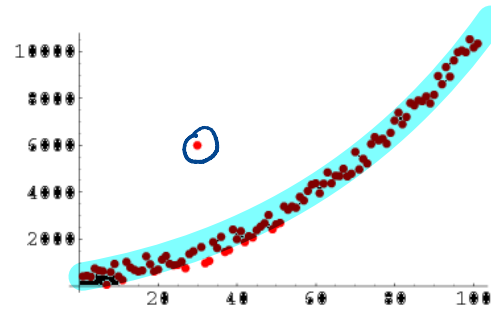
Does it look curved?

Does it look cyclical?

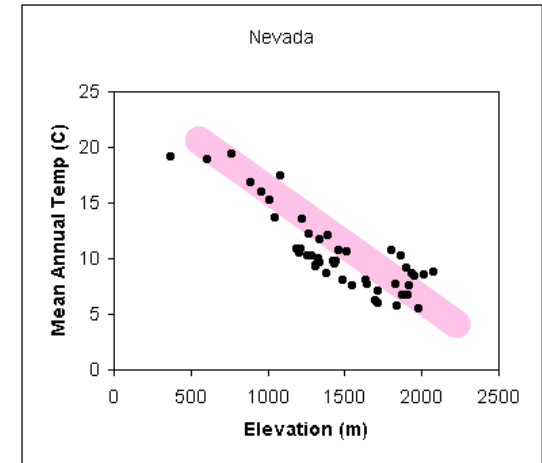
Does it look random?



- Outliers: Yes
- Clusters: No
- Association: Positive
- Functional Form: Linear

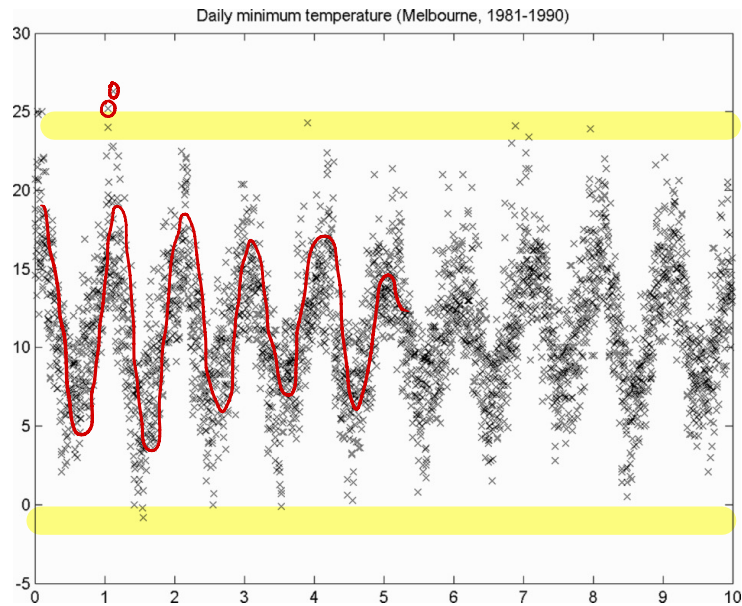


- Outliers: Yes
- Clusters: No
- Association: Positive
- Functional Form: Curved

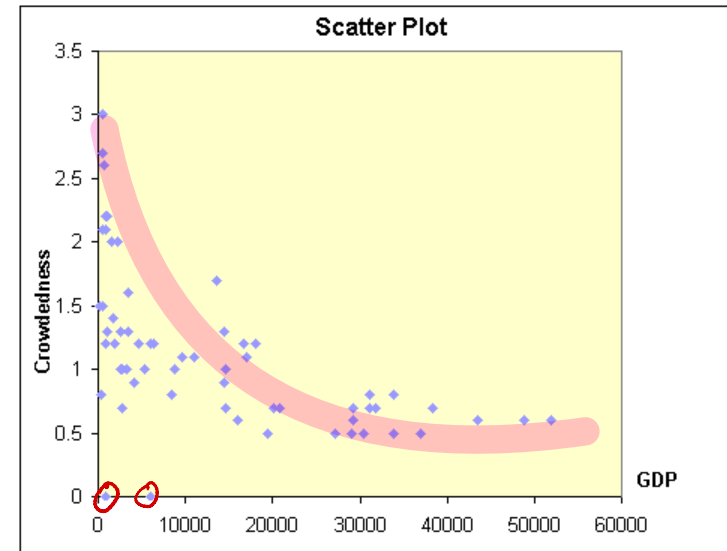


- Outliers: No
- Clusters: No
- Association: Negative
- Functional Form: Linear

Examples:

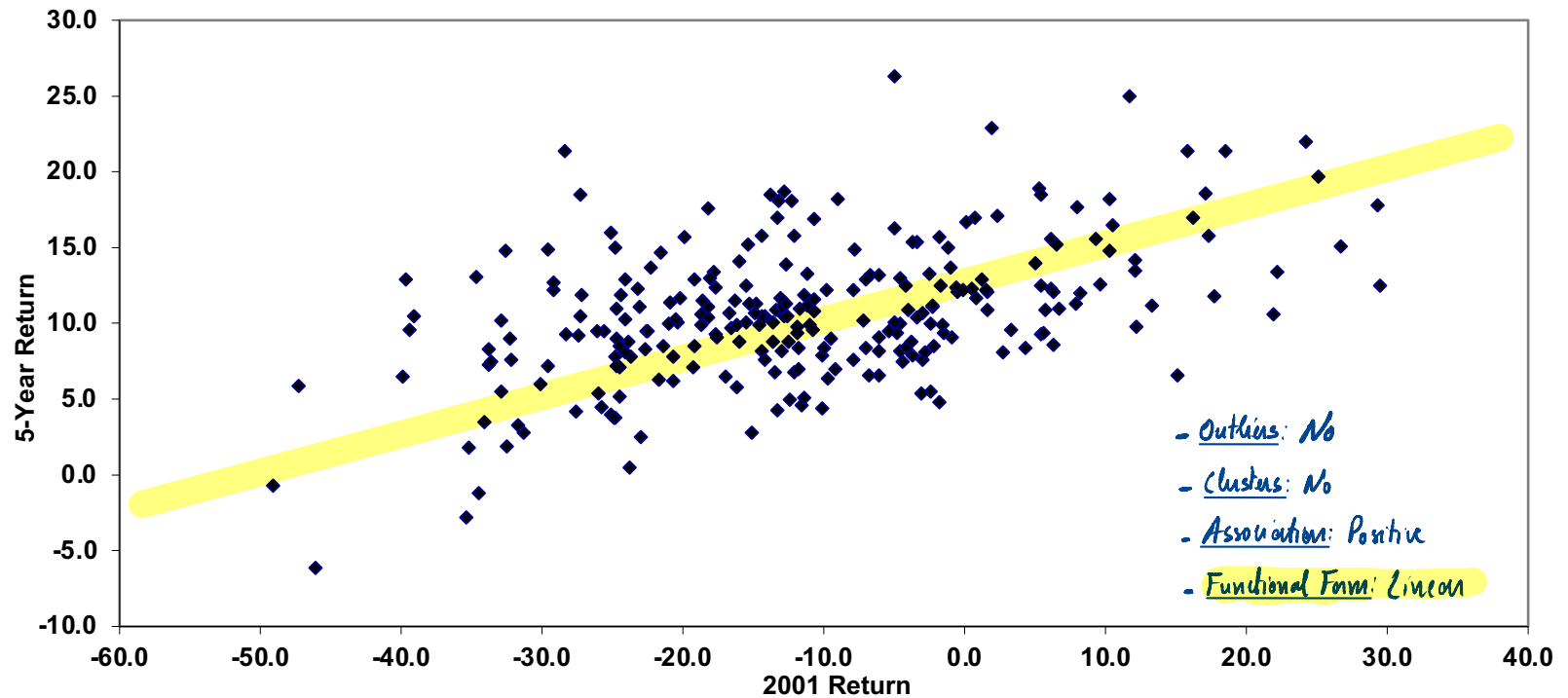


- Outliers: Yes
- Clusters: No
- Functional Form: Cyclical



- Outliers: Yes
- Clusters: No
- Association: Negative
- Functional Form: Curved

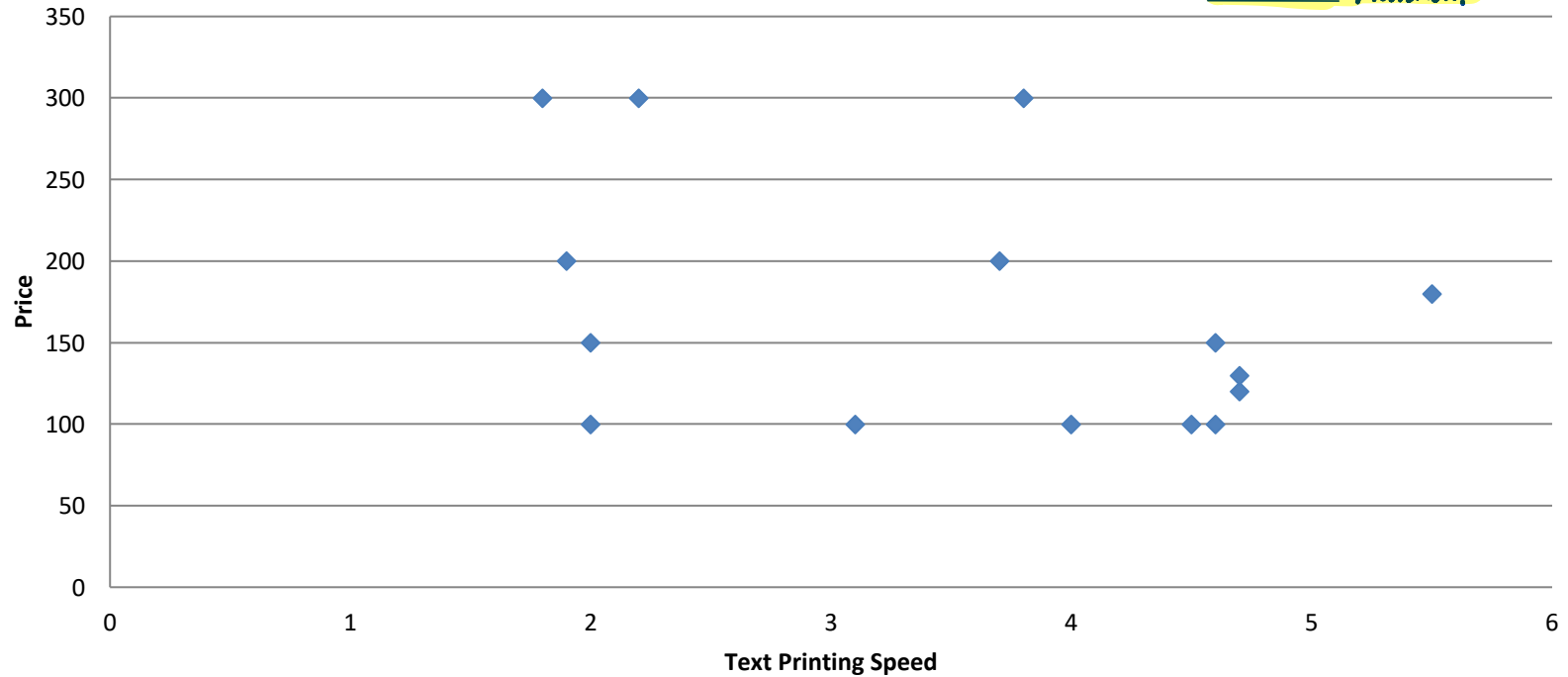
Mutual Fund Returns



- **Source:** *Statistics for Managers Using Microsoft Excel 4th Edition*, Levine, Stephan,
- Krehbiel, Berenson, Figure 2.5 on page 65

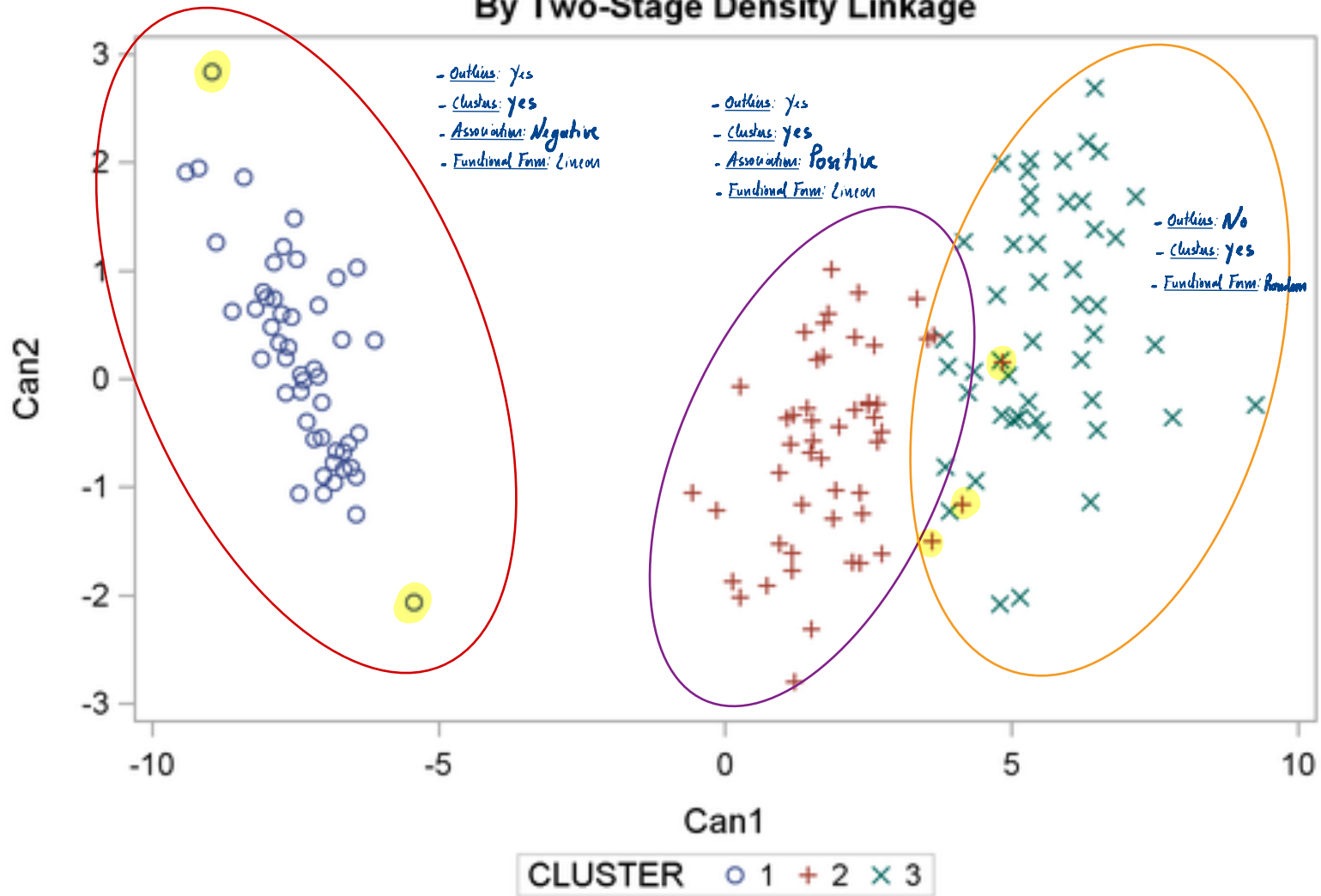
Printer Cost

- Functional Form: Random



- **Source:** *Statistics for Managers Using Microsoft Excel 4th Edition*, Levine, Stephan,
- Krehbiel, Berenson, PRINTERS.xls data on p. 88

Cluster Analysis of Fisher (1936) Iris Data By Two-Stage Density Linkage





Reading

- CHAPTER 2 – SECTION 2.1

Next Topic



Chapter 2 – Section 2.1



Correlation